Technical & Strategic Documentation: Property Finder RecSys (v12.4-Production)

1. Executive Summary

The **Property Finder Recommendation System (v12.4-Production)** is a sophisticated, high-performance intelligence layer designed to evolve the property search experience from a static database query into a dynamic, intent-driven journey.

By integrating **XGBoost gradient boosting** with **NLP-based behavioral profiling**, the engine delivers real-time, hyper-personalized rankings. It processes an inventory of **44,000+ listings** against a historical dataset of **1.55 million user interactions**, ensuring that every result is mathematically optimized for both market quality and individual user taste.

2. Core Philosophy: "Intent over Attribute"

Traditional real estate platforms rely on "Hard Filtering" (e.g., "Show me 2-bedroom apartments"). Our philosophy, refined in version 12.4, recognizes that user behavior is a more accurate signal of desire than UI selections. The system operates on three psychological layers:

Layer 1: The Rational Boundary (Constraint Satisfaction)

We respect the user's non-negotiable boundaries, such as budget, location, and property type. These act as "Hard Gates," ensuring the engine remains grounded in the user's practical reality.

Layer 2: The Social Proof (Market Quality)

Trust is the primary currency in real estate. This layer uses global machine learning models to identify "Market Winners"—listings that are verified, competitively priced, and managed by top-tier "Super Agents." This ensures that even without user history, the platform surfaces the highest-quality inventory first.

Layer 3: The Subconscious Affinity (Lifestyle Matching)

Users often have a "vibe" or aesthetic preference they cannot articulate. By analyzing historical interaction "DNA" (e.g., a pattern of clicking properties with high-floor views or private pools), the engine builds a **Subconscious Profile**. It surfaces properties that match the user's "soul," even if they fall outside standard sorting patterns.

3. Technical Methodology

3.1. Data Infrastructure

The engine utilizes a **Parquet-First** architecture, chosen for its high-speed I/O capabilities and efficient handling of large-scale datasets.

- **Inventory Dataset:** 44,127 active listings with 38 feature columns.
- **Interaction Dataset:** 1,555,000 user interaction rows (clicks/leads).
- **Synchronization Logic:** A strict "Type-Safe Bridge" ensures that `property_listing_id` fields are standardized across legacy history and active snapshots to prevent data drift.

3.2. The Hybrid Scoring Formula

The final ranking of a property is determined by a multi-variate mathematical model:

$$RankScore = XGBoost(MarketQuality) + (Similarity(UserDNA, PropertyDNA) \times Weight)$$

A. Static ML Layer: XGBoost (brain.pkl)

We deployed a Gradient Boosting Regressor trained on historical performance metrics.

- **Input Features:** `price`, `size_sqft`, `super_agent_score`, `verified_flag`, and `listing_level`.
- **Objective:** To predict the objective "Market Value" of a listing.

B. Dynamic NLP Layer: Vectorization & Similarity

To achieve personalization, we treat property amenities as a language.

- **Tokenization:** A custom `amenity_tokenizer` parses codes (e.g., `BA` for Balcony, `VW` for View).
- **Centroid Calculation:** We aggregate the vectors of every property a user has interacted with to create a **"User DNA Centroid."**
- **Cosine Similarity:** We calculate the mathematical distance between the User Centroid and the Candidate Property. If a match is found (e.g., both share a "Waterfront" tag), a **5.0x Linear Boost** is applied to the property's rank.

4. Key Functional Features

Feature	Technical Execution	Business Value
Radius Geospatial	Haversine Vectorization Formula	Allows "Search Near Me" with kilometer-perfect precision.
Commercial Logic	ID-conditional Formatting	Correctly labels "0 Bedrooms" as "Land/Office" for Commercial, but "Studio" for Residential.
Amenity Intersection	Boolean Array Masking	Ensures strict "AND" logic for selected amenities (e.g., Pool AND Gym).
Verified Priority	Weight-Biased ML	Increases platform trust by prioritizing "Verified" and "Premium" listings.
Real-time Re-ranking	Asynchronous FastAPI Search	Delivers personalized results in <100ms.

5. Diagnostic Verification & QA

During the final production audit, the system underwent rigorous testing to ensure logic integrity:

1. **Schema Audit:** We identified and resolved a data-type mismatch (Integer vs. String) that was causing interaction history to be ignored. The "Force-String Bridge" now ensures 100% connectivity.
2. **The "Luxury Shift" Test:** We successfully simulated a high-intent user (interested in Waterfront/Luxury). The engine correctly promoted a **$52M Luxury Listing** to the #1 spot, leapfrogging cheaper but less relevant properties.
3. **Graceful Degradation:** For new users (Cold Start), the system automatically pivots to Layer 2 (Market Quality), ensuring a premium experience even without historical data.

6. Implementation & Scalability

The system is built for the modern web:

- **Backend:** FastAPI (Python) for high-concurrency handling.
- **Data Science:** Scikit-Learn, Pandas, NumPy, and XGBoost.
- **API Contract:** Returns standardized JSON objects containing `meta_data`, `rank_score`, and `ProtoProperty` objects for seamless UI integration.
- **Deployment:** Stateless architecture, fully compatible with Docker and AWS/GCP cloud environments.

7. Conclusion

The **v12.4-Production** engine is not merely a filter—it is a **Conversion Engine**. By understanding the subtle patterns of human behavior and combining them with rigorous market analysis, the system reduces "search fatigue" and accelerates the journey from a click to a contract.

Status: Production Ready **Version:** 12.4.0 **Lead Logic:** Hybrid XGBoost/NLP Personalization